

Hamza Aziz

AI Engineer | Generative AI & Computer Vision

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years building production AI that is safe to run on real customers: voice agents, LLM systems and GPU computer vision. Helped lift live drive-thru ordering success to 60%+; built face recognition on 20+ live CCTV streams at >99.99% accuracy in controlled UAT; shipped 2 live products.

SKILLS

- **Generative & Conversational AI:** Google Gemini, Vertex AI, LangGraph, LangChain, RAG, vLLM, Rasa, Prompt Engineering
- **ML, Computer Vision & Edge:** PyTorch, TensorFlow, OpenCV, YOLO, InsightFace, TensorRT, ONNX, DeepStream, Jetson
- **Backend & Data:** Python, SQL, FastAPI, Socket.IO, PostgreSQL, MongoDB, Redis, Milvus, Kafka
- **DevOps & Infrastructure:** Docker, CI/CD, Git, Linux, MinIO, DGX A100 (MIG)

WORK EXPERIENCE

AI Product Engineer — *VoicePlug Inc.* | September 2025 – Present

- Helped lift live client success from ~25–30% to 60%+ at 2 live restaurants (4+ QSR brands) by fixing failures found in 500+ real customer interactions
- Kept pricing and order finalisation out of the LLM: one cached Gemini 2.5 Flash call inside deterministic guards, with application code owning pricing and commit across 4 POS integrations
- Bounded cost and latency with per-store Vertex AI context caching, Flash/Flash-Lite tiering across 5 model roles and a 1,024-token reasoning cap: 2.63 s median reply over 17,884 production turns
- Built a replay harness running each case 3× for model variability and an ops loop from production logs to fixes, reducing recurring production issues by 6%

Assistant Engineer – AI — *Global InfoVentures Pvt. Ltd.* | August 2023 – August 2025

- Built CCTV face-recognition attendance as sole developer (InsightFace-REST, TensorRT, Milvus): >99.99% accuracy in controlled UAT; 4 CCTV, PTZ and edge variants on DGX A100, RTX 4090 and Jetson Orin Nano
- Processed 20+ live CCTV streams in 32-frame batches across 4 recognition threads on A100 MIG slices, for a 20-camera main gate averaging ~2,000 people a day
- Lifted real-world accuracy from 85% to 99% with nightly gallery updates; mobile + CCTV training data improved cross-domain recognition by 14%
- Built a PTZ classroom-attendance system as sole developer: 23 ONVIF presets per camera, YOLOv9 headcounts and Milvus matching, with ERP exports every 5 minutes
- Worked in the team that built G6, a voice ERP assistant for 4 institutes; schema-grounded query generation improved SQL query precision by over 30%

AI Intern — *Global InfoVentures Pvt. Ltd.* | February 2023 – July 2023

- Delivered 10+ AI/ML training sessions, mentored 50+ students and guided 10+ YOLOv8/CNN projects

PROJECTS

Codex–Claude Council (npm, MCP Registry)

- Open-source MCP server where ChatGPT and Claude debate until they agree; 118 tests on 3-OS CI; in the MCP Registry and awesome-mcp-servers

Jetson Nano Face Recognition Attendance System (DeepStream, TensorRT)

- Moved face detection into an adaptive pad probe on a Jetson Nano 4 GB DeepStream pipeline (SCRFD + ArcFace, TensorRT FP16): idle GPU load from 89–99% to near zero, 12.2 → 15.0 fps on a 1080p camera

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani – WILP — 2025 – 2027
Master of Technology (Executive), Artificial Intelligence and Machine Learning — in progress

ABES Institute of Technology, AKTU — Computer Science and Engineering
Bachelor of Technology — 2019 – 2023

CERTIFICATES

Rasa Developer (2025) · Generative AI with LLMs, Coursera (2024) · NVIDIA DLI Jetson Nano (2024)